

CHAPTER 10

Quantum Localization in a Broader Conceptual Perspective

10.1 Boolean Covers and Physical Contexts

The association of a *local Boolean cover* in the mathematical descriptive language of sheaves, with a concrete *physical context* in the physical world, takes into account two fundamental distinctions inextricably connected with the quantum-theoretic formalism: The first of them refers to a distinction being made between an *event* and the *physical context* that constitutes a set of necessary and sufficient constraints for the occurrence of an event of the observed kind. To the event, there corresponds a formal descriptive proposition language. To the physical context there corresponds a context-description in a formal descriptive language assuming existence at the level of Boolean covers, followed by an appropriate terminology providing names for the characterization of the language of events occurring in that context. These latter descriptions can be said to belong to the constitutive level of the Boolean localization systems. The second refers to a distinction being made between *possessed* physical quantities, as those found in classical physics, and *dispositional* ones, as those found in quantum physics. The dispositional character of quantum observables is associated with the fact that they may only be specified via the measurement process, and more precisely, as *relationally* appearing with respect to theoretical or actual Boolean preparatory environments, that is, Boolean logical frames. In the mathematical descriptive terminology this distinction is encoded in a *transition* from globally Boolean event structures to globally non-Boolean event structures, being *covered* by epimorphic families of local Boolean reference frames forming covering sieves. In this sense, creating a preparatory Boolean environment for a system to interact with a measuring device *does not* determine which event will take place, but it *does* determine the *kind* of event that will take place. It forces the outcome, whatever it is, to belong to a certain definite *Boolean algebra* of events for which the standard measurement conditions

are invariant. Such a set of standard conditions for a definite kind of measurement is named a physical context and reflects a Boolean reference frame in the mathematical descriptive language.

10.2 Partial Congruence and Adjunction

From a general category-theoretic perspective the *relational realist* framework proposed for the interpretation of quantum event structures and their logical semantics is based on the existence of the Boolean frames-quantum adjunction. From this adjunction, characterized by means of the counit and unit natural transformations, we have constructed a *sheaf-theoretic representation* of quantum event algebras formulated with respect to Boolean localization systems, as well as a quantum subobject classifier, which plays the role of a *logical classifying object* in the quantum universe of discourse. In this way, the Boolean frames-quantum adjunction incorporates both the *representational semantics* of quantum event algebras as sheaves of local Boolean coefficients, and the *logical semantics* of the classifying Boolean germs concerning truth valuations with respect to Boolean localization systems.

Thus, from a physical viewpoint, the Boolean frames-quantum adjunction stands as a theoretical platform for *decoding* the global structural information contained in quantum algebras of events via *processes of localization* with respect to Boolean logical frames, realized as physical contexts for measurement of observables, and subsequent *processes of information classification* in terms of corresponding truth valuations. The functioning of this platform is based on the establishment of a *bidirectional* dependence between the Boolean and quantum structural levels being brought into *local* or *partial* congruence. Most significantly, this bidirectional dependence is implicated through the topos-theoretic universe of sheaves of sets over local Boolean frames of quantum events algebras forming Boolean localization systems, where these local Boolean frames play the role of local Boolean covers, effectuating in this way the specific form of the above partial congruence. Of course, the sheaf-theoretic requirements secure the *compatibility* of the locally Booleanized information in the overlapping regions of physical measurement contexts under extension from the local to the global and conversely.

Additionally the *global closure* of this bidirectional dependence is necessary to be constrained in order to obey certain conditions, such that its total constitutive information content, unfolded in the multitude of local Boolean reference frames, is both preserved and coherently organized in a logical manner. Remarkably, the necessary and sufficient conditions for both of these requirements, that is: [i] *preservation* of quantum information content in Boolean localization systems associated with compatible physical measurement contexts; and [ii] *logical classification* of quantum information by means of the

classifying Boolean germ *true* (equivalence class) used for the valuation of quantum propositions, are supplied by the Boolean frames-quantum adjunctive correspondence itself, via the counit and unit natural transformations, respectively.

More concretely, regarding the first condition, we conclude that it is satisfied if the *counit* of the adjunction is an *isomorphism* for each quantum event algebra. In this case, there exists a full and faithful sheaf-theoretic representation of quantum events algebras in the descriptive terms of Boolean covering systems, characterized as Boolean localization systems of measurement. Regarding the second condition, we conclude that it is satisfied, given the *validity* of the first condition, if the *unit* of the adjunction is an *isomorphism* for the subobject functor. In this case respectively, the subobject functor is representable in the category of quantum event algebras by a *classifying object* in this category, called a *subobject classifier*. This classifying object, which is defined by an *inductive limit* operation, contains the information of equivalence classes of truth valuations with respect to all compatible Boolean frames belonging to a Boolean localization system of a quantum event algebra L in $\mathbf{L}$. Thus, it can be legitimately used for *truth valuations* of quantum propositions in exact correspondence with the use of the two-valued Boolean object, which is used for valuations of classical propositions. Moreover, the quantum subobject classifier provides the key logical device for the analysis of typical quantum measurement situations, providing a *criterion of truth* for a complete description of reality in the quantum regime with respect to a Boolean localization system.

10.3 Classical to Quantum:

From Groups to Categories

It is instructive to draw an *analogy* between the notion of a reference frame as it is used in classical physics (presupposing the existence of a set-theoretic *spacetime* substratum) and the notion of a variable *Boolean localizing reference frame* in quantum physics. In classical or relativistic physics we always have to define first the notion of a reference frame (spatiotemporal reference frame) where we may describe a physical phenomenon or formulate a physical law. The consideration of a single reference frame is not enough. What we have to consider is a whole *class* of reference frames and their *transformations*. In classical theories the class of all reference frames together with their interrelations form a *group* of transformations. Then the physical laws involved should be *invariant* under the action of this group of transformations.

Now, a group of transformations is a very special kind of category having only one object. It is a natural requirement, in case we generalize to quantum theory for example, to obtain an analogous

description of phenomena. For this purpose, we generalize from a spatiotemporal reference frame (*a metrical notion*) to an analogous localizing notion of reference frame (*a topological notion*), which admits a concrete physical interpretation and plays a similar role. This is the functionality of a Boolean reference frame in quantum theory. Thinking categorically means that we have to specify the transformations from a Boolean reference frame (corresponding to a quantum measurement situation) into another. Now the class of all Boolean reference frames and their transformations do not form a group but they form a *category*. So the quantum description of phenomena should be *invariant* under the action of the concrete category of all partially compatible Boolean reference frames. More concretely, the physical content of the topological sheaf-theoretic representation of quantum event algebras can be formulated in terms of an invariance principle. According to this, *the information content of a quantum event algebra is invariant under the category of gluing isomorphisms (technically called a groupoid) between overlapping local Boolean reference frames, along their intersections, in a Boolean localization system, preserving the quantum algebraic structure.*¹

It is important to notice the obvious analogy to the metrical spacetime frame case. The requirement of *invariance* is the conceptual reason necessitating the introduction of the notion of a sheaf of Boolean logical frames, or equivalently the notion of a Boolean topological localization system of a quantum event algebra. We emphasize that the normative requirements for the localization of information in the quantum case *does not* refer to set-theoretic spacetime localization. Categorically thinking, localization has to do with the partiality of the information referring to a Boolean reference frame, interpreted in topological terms, and its consistent gluing effectuating a coherent *extension* or *induction* from the local to the global level.

It is instructive to note that in the special case of a group of transformations (that is the usual spacetime case with point-defined spatiotemporal reference frames) the notion of categorical localization is identical with the notion of spacetime localization. So, using the categorical framework we incorporate the classical spacetime localization case as a *very special case of the general localization notion* (the former is referring to a group of transformations, whereas the latter is referring to a category of transformations among the corresponding frames). Presheaves and sheaves (incorporating the gluing compatibility conditions) provide the conceptual and technical devices in order to express such a *generalization in the notion of localization*, which finds an important application in quantum theory (by introducing Boolean reference frames). Now the *gluing process* is a *vast generalization* of the usual process of requiring *inter-transformability* of our descriptions when we pass from some reference frame to another, if we take into account this more general localization idea. The benefit is that we obtain additionally a coherent *mereotopological* semantic methodology of extension from the local or partial

into the global (considered in observable-theoretic terms as it is actually practiced via physical measurement) and conversely.

We claim that it is precisely the *sheaf-theoretic gluing process* which provides a rich and adequate mathematical model of Whitehead's scheme of *extensive connection*. The distinguishing feature of this claim is that the gluing process is of a purely *topological* nature and thus it captures the character of the relationship of extension idealized by Whitehead as being *prior to the metrical notions* of space and time:

In this general description of the states of extension, nothing has been said about physical time or physical space, or of the more general notion of creative advance. These are notions which presuppose the more general relationship of extension. They express additional facts about the actual occasions. The extensiveness of space is really the spatialization of extension; and the extensiveness of time is really the temporalization of extension. Physical time expresses the reflection of genetic divisibility into coordinate divisibility. . . ²

10.4 Topological Localization as Genetic Analysis

From a general philosophical standpoint, the general conceptual underpinning of the notion of sheaf-theoretic localization of observability is deeply related to the theoretical systems of Aristotle and Whitehead. According to Aristotle the *divisions* that we project into an observable continuum assume a *potential* existence, unless their *actuality* can be confirmed by property-indicating facts, viz., by experimental events. Thus, from an Aristotelian perspective, ontological *primacy* should be shifted from the level of states, to the level of observable and event information structures. This means that any complex systems theory (including quantum theory) should be viewed as a theory specifying a particular way of *correlating property-indicating events*, according to a schema of *potential division* of an observable continuum in terms of *local information carriers* (for instance Boolean reference frames), assuming its actuality *only* by the observed events and their correlations.

The process-theoretic system of Whitehead advocates this conception and poses additionally the natural condition that the correlations should conform to a *theory of congruence*, which should act as a normative requirement in order to make sense of the observed events and their correlations. Furthermore, a theory of congruence would manifest the potential division of an observable continuum as commonly presupposed in the study of natural phenomena, whereas its functional necessity and essential correctness would be affirmed only by means of the observed events (*actual occasions*).

From a logical perspective, observed events in quantum theory correspond to evaluations of potential facts (observables) with respect to some logical binary schemata (yes/no experiments subsumed by

measuring devices), interpreted as property-indicating facts. Note that this is meaningful from a topological perspective only if it respects the normative constraints of the potential division of a global quantum event continuum with respect to local Boolean reference frames (preparation stage of a quantum measurement), as well as the rules of compatibility under extension from the local to the global and conversely. The *local-to-global bidirectional scheme* corresponds to Whitehead's notion of *genetic analysis*, and is reflected in the modeling of the continuum related to the behavior of a quantum system via a *category of sheaves* of local information carriers (local Boolean reference frames). More precisely, the action of a presheaf corresponds to a potential division of the continuum according to the normative rules of a congruence theory of local Boolean observables, whereas the transformation of a presheaf into a sheaf (that is to a Boolean localization system) reflects the gluing and compatibility conditions determining the correlations of observed events under the transition from the local to the global, and conversely under the reduction of the global to the local.

Note that in full agreement with Whitehead, the genetic passage from phase to phase, viz., from each local Boolean observable frame to another, is *not* in physical time, meaning that the genetic process is *not* the temporal succession. *This is due to the dependence of each local phase in the genetic process on the global level, meaning that each such phase presupposes the global quantum level implicitly.* It is also important to notice that the conditioning of actualized events by such phases (considered as local Boolean reference frames) is indispensable for the existence of properties. In other words, every property is a possessed property, indicated by an actualized event, which is conditioned on the existence of a local phase, presupposing the global quantum level.

10.5 Boolean Localization and Decoherence

A significant application of the proposed Boolean localization scheme of quantum event structures concerns the elucidation of the phenomenon of quantum decoherence.³ *Decoherence* is understood as the phenomenon according to which all the features of quantum observable behavior are suppressed, and instead a quantum system *behaves like* it can be described by classical probabilistic reasoning. Typical quantum features of this form constitute interference phases and the existence of off-diagonal elements in the density matrix description of a quantum system. Thus, decoherence is mathematically modeled either by the process of *diagonalization* of the density matrix with respect to a particular basis, called technically the *pointer basis*, or by the process of *minimization of interference phases* in a sufficiently *coarse-grained* resolution scale of quantum observable behavior. The essential aspect of all modeling approaches to decoherence is the explanation of the *emergence of classical behavior*, meaning the emergence of a classical statistical description referring to ensembles of equivalently prepared systems.

The first of the current modeling approaches to decoherence may be called *environment induced* decoherence, whereas the second may be called *coarse-graining* induced decoherence. These approaches are based on different conceptual foundations regarding the explanation of emergence of classical behavior. More specifically, environment-induced decoherence is an *extrinsic* approach in the sense that it requires the coupling action of an external to the system agent, called the *environment*, in order to explain the emergence of a classical statistical description with respect to some particular basis, that is the pointer basis. The crucial aspects of this approach may be summarized by stressing the *contingent* and extrinsic influence of some environment on a quantum system and the conception that the process of decoherence requires the passage of time, called *decoherence time*, within which energy and information are being dissipated to the environment. On the other side, coarse-graining induced decoherence is an *intrinsic* approach in the sense that classical behavior emerges intrinsically within the description of a quantum system by *coarsening* the scale of resolution of quantum observable behavior by means of *macroscopic* or gross or collective observables, called classical observables. Hence, decoherence is not conceived as a process requiring the coupling of a quantum system to some external environment, but as a process of resolving quantum observable behavior at sufficiently coarse-grained scales, expressed in terms of *macroscopic algebras of observables*, with respect to whom interference phases are being suppressed. The crucial aspects of the latter approach may be summarized by stressing the intrinsic character of emergence of a classical statistical description in terms of *commutative algebras of macroscopic observables* incorporated at sufficiently coarse-grained levels within a *global non-commutative algebra* of microscopic quantum observables.

We argue that the full implications of coarse-graining induced decoherence require a *topological* modeling approach based on the functional role of Boolean localization systems. More concretely, the phenomenon of decoherence should be understood as a consequence of covering the intrinsic structure of a global non-commutative (non-Boolean) algebra of quantum observables by covering sieves of local commutative (Boolean) algebras of macroscopic or classical observables corresponding to suitable coarse-grained resolution scales of quantum measurement and then localizing sheaf-theoretically at these scales.⁴

The crucial mathematical advantage of this topological approach to quantum decoherence is that *the phenomenon of decoherence can be understood as a process of topological localization of a global quantum (non-Boolean) algebra of observables at some local commutative (Boolean) algebra of classical observables associated with some macroscopic resolution scale*. Topologically speaking, this becomes possible via sieving the global information content of a quantum algebra of observables with respect to compatible covering families (covering sieves) of local commutative (Boolean) algebras of

macroscopic observables assuming existence at various *coarse-grained* measurement levels. In this conceptual framework, it becomes possible to understand the notion of environment *not* as an external agent forcing the selection of some pointer basis with respect to whom a classical statistical or Boolean description emerges, but *internally as localizing* topologically a global algebra of fine-grained microscopic observables at some coarse-grained macroscopic level *sufficient* for emergence of classical behavior.

Most significantly, the topological approach not only models decoherence via the sheaf-theoretic mathematical technique of Boolean localization via coarse-graining sieves, but allows the conceptualization and modeling of *recoherence* from the local macroscopic to the global microscopic level of observable structure via sheaf-theoretic amalgamation of locally compatible observables, that is by gluing locally compatible observables together globally in the form of equivalence classes, which we have called *Boolean germs*.

The conceptual motivating precursors of the topological sheaf-theoretic modeling of coarse-graining induced decoherence are, on the physical side, the *consistent or decoherent histories* approach to quantum theory⁵ and on the mathematical side, the *topological representation theorems* of algebras of observables by *sections of a sheaf* over a topological space, or a category endowed with a Grothendieck-type of a topology, tracing their roots to the famous *Stone and Gelfand topological representation theorems* of Boolean and commutative algebras, respectively.⁶

On the physical state of affairs, the consistent histories approach to quantum theory focuses on the formulation of an appropriate criterion, subsumed in the formulation of a *complex-valued functional* for pairs of histories, the vanishing of which specifies the condition of *consistency*, that is it decides which histories decohere. Typically, *decoherent sets* of histories contain coarse-grained histories. More precisely, in quantum mechanics, propositions about the attributes of a system at a fixed moment of time are represented by sets of projection operators. The projection operators P_α enable a partition of the possible alternatives α a system may exhibit at each moment of time. They are exhaustive and exclusive,

$$\sum_{\alpha} P_{\alpha} = 1, \quad P_{\alpha} P_{\beta} = \delta_{\alpha\beta} P_{\alpha}$$

A projector is said to be *fine-grained* if it is of the form $|\alpha\rangle\langle\alpha|$, where $\{|\alpha\rangle\}$ are a complete set of states. Otherwise it is *coarse-grained*. A *homogenous quantum-mechanical history* is characterized by a

string of time-dependent projections, $P_{\alpha_1}^1(t_1), \dots, P_{\alpha_n}^n(t_n)$, together with an initial state ρ . The time-dependent projections are related to the time-independent ones by:

$$P_{\alpha_k}^k(t_k) = e^{iH(t_k-t_0)} P_{\alpha_k}^k e^{-iH(t_k-t_0)}$$

where H is the Hamiltonian. The *candidate probability* for these homogeneous histories is:

$$p(\alpha_1, \alpha_2, \dots, \alpha_n) = \text{Tr} \left(P_{\alpha_n}^n(t_n) \cdots P_{\alpha_1}^1(t_1) \rho P_{\alpha_1}^1(t_1) \cdots P_{\alpha_n}^n(t_n) \right)$$

It is straightforward to show that $p(\alpha_1, \alpha_2, \dots, \alpha_n)$ is both nonnegative and normalized to unity when summed over $\alpha_1, \dots, \alpha_n$. However, it *does not* satisfy all the axioms of probability theory, and for that reason it is referred to as a candidate probability. For example, it does not satisfy the requirement of *additivity* on disjoint regions of sample space. More precisely, for each set of histories, one may construct coarser-grained histories by grouping the histories together. This may be achieved, for example, by *summing over* the projections at each moment of time, $\bar{P}_{\bar{\alpha}} = \sum_{\alpha \in \bar{\alpha}} P_{\alpha}$. The additivity requirement is then that the probabilities for each coarser-grained history should be the sum of the probabilities of the finer-grained histories of which it is comprised. Quantum-mechanical interference generally prevents this requirement from being satisfied. Histories of closed quantum systems *cannot* in general be assigned probabilities. There are, however, certain types of histories for which interference is negligible, and the candidate probabilities for histories do satisfy the sum rules. These histories may be found using the complex-valued *decoherence functional*:

$$D(\underline{\alpha}, \underline{\alpha}') = \text{Tr} \left(P_{\alpha_n}^n(t_n) \cdots P_{\alpha_1}^1(t_1) \rho P_{\alpha'_1}^1(t_1) \cdots P_{\alpha'_n}^n(t_n) \right)$$

Here $\underline{\alpha}$ denotes the string $\alpha_1, \alpha_2, \dots, \alpha_n$. Intuitively, the decoherence functional measures the *amount of interference* between pairs of histories. It may be shown that the additivity requirement is satisfied for all coarse-grainings if and only if the *real part* of the complex-valued decoherence functional vanishes, that is $\text{Re} D(\underline{\alpha}, \underline{\alpha}') = 0$, for all distinct pairs of histories $\underline{\alpha}, \underline{\alpha}'$. Such sets of histories are said to be consistent, or weakly decoherent. The consistency condition is typically satisfied only for coarse-grained histories, and this then often leads to satisfaction of the stronger condition of decoherence $D(\underline{\alpha}, \underline{\alpha}') = 0$ for $\underline{\alpha} \neq \underline{\alpha}'$. The condition of decoherence is associated with the existence of so-called *generalized records*, corresponding to the idea that information about the variables to follow is stored in the variables ignored in the coarse-graining procedure.

We may state concisely the key idea of the consistent histories approach to quantum mechanics by the realization that we can have additive probabilities only in case we *restrict* our resolution scale within

particular sets of histories, called *consistent sets*, which are typically sets of appropriately coarse-grained histories. The main problems of the consistent-histories approach to coarse-graining induced decoherence may be summarized as follows:

- [i] It uses an *external time* parameter to both formulate the notion of history, and express the process of coarse-graining by summation of projection operators at each moment of this external time;
- [ii] It turns out that there exist *many consistent sets* of histories which are mutually *incompatible* such that they cannot be combined to give a maximal consistent set, and at the same time there is *no criterion* of selecting uniquely some consistent set among them; and
- [iii] The criterion of consistency of an arbitrary set of histories is *not correlated* with the requirement of classical behavior of a large class of observables. This is due to the fact that the difference between the algebraic structure of quantum observables (non-Boolean and non-commutative) in comparison to the algebraic structure of macroscopic observables at suitable coarse-grained scales (Boolean and commutative) is essentially *ignored* in this formalism.

In order to cope with these problems and formulate a more satisfactory account of coarse-graining induced decoherence, we need to topologize quantum algebras of observables in the spirit of Stone and Gelfand topological representation theorems of Boolean and commutative algebras respectively.⁷

The link between the consistent histories approach and the topological approach is obtained in two steps:

- [i] By considering explicitly the *difference in the algebraic structure* between an algebra of microscopic observables and various algebras of macroscopic observables at suitable coarse-grained measurement scales with respect to the property of *commutativity*, and
- [ii] By *topologizing* this difference in terms of a local-to-global sheaf-theoretic construction, where the information of a global non-commutative algebra of quantum observables may be *localized* by *restriction* or *reduction* to *compatible families* of local commutative algebras of classical macroscopic observables topologically *covering* the former, and inversely, recovering its content by the topological method of *gluing*.

At the propositional level, that is at the level of projection operators, as we have already pointed out, it is a common realization that existing set-theoretic lattice axiomatizations of quantum observable structures hide the intrinsic fundamental significance of Boolean commutative localization systems in the formation of these structures. On the other side, topologically, we can explicitly associate them with the action of appropriate Boolean coverings of global quantum algebras. Consequently, the algebraic representation of a quantum observable structure in the relational topological terms of localization systems of commutative algebras of macroscopic Boolean observables, effectuates a *semantic transition*

from the axiomatic set-theoretic context of orthocomplemented partially ordered sets, *à la* Birkhoff and Von Neumann, to the categorical sheaf-theoretic context of Boolean information sieves, *à la* Grothendieck.

In order to understand the semantic transition of quantum observable algebras from the set-theoretic level to the sheaf-theoretic level, as well as the significance of this transition in relation to the conceptual and technical modeling of decoherence, it is instructive to explain briefly the philosophy behind Stone and Gelfand topological representation theorems of Boolean and commutative algebras, respectively, being the precursors of Grothendieck-type sheaf-theoretic formulations expressed in terms of topological covering families (covering sieves) on categories. These theorems state that an arbitrary commutative algebra of observables A of a certain type (for instance a Boolean algebra of observables in Stone's case) may be represented as the *algebra of continuous F -valued functions*, where F is a particular *standard algebra* of the given type, on a *topological space* associated with A and constituting in this way its F -*spectrum*. For example, in Stone's case, the standard algebra is the two-valued Boolean algebra $\{0,1\}$ and we make use of the discrete topology. The topological representation theorem is proved by the fact that a continuous function from the topological $\{0,1\}$ -spectrum space X to $\{0,1\}$, that is $f:X \rightarrow \{0,1\}$ is completely determined by the clopen subset $f^{-1}(1)$ of X .

Now, the extension of this topological representation philosophy to larger classes of algebras of observables proves to be more complicated in the sense that a general algebra of observables A *cannot* be built up in this way from a single standard algebra like the two-valued Boolean algebra $\{0,1\}$ in Stone's case. This means that its spectrum is more complicated. An algebra of quantum observables falls into the latter category, and this is precisely the meaning of Kochen-Specker theorem, which essentially states that it is impossible to recover the global information content of a global quantum observable algebra using a single type of a Boolean measurement device globally. Of course, this can be done *locally at a certain coarse-grained resolution measurement scale*, but the global information content may be approximated only by employing simultaneously an appropriate *multitude of compatible Boolean devices* at different overlapping measurement scales *covering* its behavior completely. Mathematically, the inability to build up a global algebra of observables from a single standard algebra necessitates the introduction of *continuously variable* local standard algebras being capable of covering it completely. This pictorially means that we may identify the observables of A locally with *sections of a sheaf* defined on a spectrum X and taking values in one of a given class of standard local algebras, provided that the local algebra in which an observable takes values may vary continuously from point to point in the spectrum X .

The extrapolation of the sheaf-theoretic topological representation methodology, in the case of a quantum algebra of observables, sheds new light on the process of coarse-graining induced decoherence and the emergence of classical behavior. *In a nutshell it implies that quantum observables are not specified globally by their measured values, but they are specified by their Boolean observable germs*, viz., equivalence classes of locally compatible Boolean observables forming an inductive limit in a sheaf of macroscopic coarse-grained Boolean observable algebras covering the former at various resolution measurement scales. It is instructive to stress again that the concept of a categorical topology epitomizes the meaning of a *localization scheme* of a quantum algebra, whereas the covering sieves are interpreted as generalized measures of topological localization of quantum observables at various coarsegrained macroscopic (Boolean) levels. In the same vein of ideas the notion of a site generalizes the notion of a topological state space in categorical terms and constitutes the *relational variable topological background* for the effectuation of a localization scheme for coarse-graining the global information contained in a quantum algebra through localizing Boolean algebraic frames of macroscopic observables at various coarse-grained resolution scales. Thus, from the physical point of view, covering sieves of a global quantum algebra by local Boolean frames effectuate Boolean localization systems of the former. It is instructive to note that covering sieves of a quantum algebra L are actually subfunctors of the functor of Boolean frames of L , viz., *relational covering networks* of morphisms from local Boolean frames to L , expressing the generic morphology of the *internal mereotopological relations* constituting the process-theoretic scheme of extensive connection from the variable Boolean local to the quantum global level. These subfunctors are *partially ordered* by inclusion. The partial order relation among covering sieves of a quantum algebra L is interpreted as the *operation of coarse-graining* among Boolean localization systems. A Boolean localization system forces a transition in the semantics of quantum observables from the set-theoretic to the topological sheaf-theoretic level. The fact that the functor $\mathbf{R}_L$ of Boolean frames of L is a sheaf, means that there exists a gluing compatibility isomorphism of local Boolean frames on their overlap, meaning that these Boolean frames of L cover the same observable information content of a quantum algebra L in a *compatible* way.

The topological compatibility relation between local Boolean frames of a global quantum algebra sheds new light and forces a *strict criterion of consistency* in the process of coarse-graining as it is currently implemented in the context of the consistent histories approach. More concretely, as we have already mentioned, the major problem of this approach is that the criterion of consistency of an arbitrary set of histories is not correlated with the requirement of classical behavior of a large class of observables. This problem is due to the fact that the process of coarse-graining as it is currently implemented *does not have to satisfy any constraint*, it is just a summation over projection operators

which describes the transition from a fine-grained description to some coarsegrained description. We claim that an appropriate compatibility relation has to be imposed on the process of coarse-graining projection operators in order to obtain meaningful probabilities from the decoherence functional, and thus to explain the emergence of classical behavior at suitable coarse-grained macroscopic scales. The Boolean localization scheme implies that the coarse-graining process should respect the local Boolean structural rule as we pass from a local Boolean context to another; that is, it should respect the morphology of internal relation between local Boolean frames covering a global quantum algebra.

In order to express this formally we need a compatibility condition which is given precisely by the transition morphisms from a local Boolean frame to another local Boolean frame as prescribed by the specification of the category of elements of the functor (sheaf) of Boolean frames of a quantum algebra. For example, referring to the same observable (for example, position) the proper coarse-graining procedure corresponds to a nesting of local Boolean frames, such that there exists compatibility on their overlaps preserving the local Boolean structural rule. In case we consider different observables and apply the coarse-graining process properly, we require partial compatibility on their overlapping coarse-grained Boolean frames (for instance, partial Boolean compatibility of coarse-grained position and momentum up to the limit of Heisenberg uncertainty relations) which is given by the topological gluing isomorphism. Then the formation of the inductive limit (which gives the transition from the local to the global) requires the formation of a set of equivalence classes (Boolean germs) of partially compatible Boolean observable information on all possible overlaps, glued topologically together in this way. This sheaf-theoretic methodology, based on the formation of Boolean localization systems of a quantum algebra, resolves completely the problem associated with coarse-graining in the consistent histories approach via the *idea of gluing on overlaps* such that the local Boolean structure rule is preserved via these transitions.

We may summarize the basic idea of the topological modeling approach to coarse-graining induced decoherence as follows: Although the global description of a quantum system should obey the laws of quantum logic and quantum probability, there exist coarse-grained descriptions at macroscopic levels, formulated for example in terms of *macroscopic hydrodynamic variables*, and expressed in terms of local Boolean subalgebras of a global non-Boolean quantum algebra, such that the attribution of properties by measurement valuations, effectuated by means of various distinct binary devices at these coarse-grained levels, takes place locally according to the laws of Boolean logic and classical probability theory. Essentially this means that a *local Boolean restriction or reduction of a quantum system with respect to a local Boolean context, or more generally with respect to a Boolean localization system, is conceived as a result of a decoherence process assuming validity with respect to an appropriate*

corresponding notion of coarse-graining subsumed by the function of covering sieves. Only in this case, it is then meaningful to infer predictions about the behavior of the localized quantum system, via its local macroscopic reduction with respect to some local Boolean context in a Boolean localization system obtained by topological coarse-graining, using the methodology of classical probability theory.

The crucial point is that according to the general conceptual framework of topological localization, *decoherence makes sense only as an intrinsic, locally conceived, reductive process of the global-level description of a quantum system.* Now, recent experimental work seems to verify the above theoretical perspective.⁸ More specifically, within the descriptive context of the environment-induced decoherence theories, it was assumed until now that the environment modeled by means of a bath of oscillators acts externally to a quantum system via an appropriate interaction coupling, and most importantly extends globally. These recent experiments show *on the contrary* that decoherence is caused by entities considered within a *localized environment*, and thus the global oscillator-bath picture is not appropriate to describe the phenomenon. More specifically, the role of such a locally conceived environment may be played by a nuclear spin bath, which gets internally entangled with a qubit (for example, a magnetically polarized molecule) and causing *intrinsic decoherence* without dissipation of energy.

10.6 Sheaf-Theoretic Quantum Relational

Realism

The idea of a reality admitting *objective* existence is the starting point of our scientific investigations and the conceptual basis pertaining to the realistic interpretation of physics. We argue that our proposed sheaf-theoretic modeling scheme provides a *realistic* perspective on quantum structure, although *conceptually different* from the classical one. The classical view of reality presupposes that objects of our observation are simply individual entities in the world. Moreover an entity may be specified, both qualitatively and quantitatively, in a definite manner *independently* of procedures of observation, being capable of assuming individuality in isolation. According to the interpretative standards of classical realism, an objective description is one that determines the properties possessed by an independently real physical object, standing for an entity, by adopting a representation of that object as a physical system isolated from any observational interaction. Quantum theory does *not* conform to the descriptive ideals of classical realism. Thus, the proposed sheaf-theoretic localization scheme of quantum event algebras should *not* be judged by the descriptive ideals of the classical realism position, but instead should be considered as generating a generalization of the classical framework, mainly by initiating a *revision* of classical realism assumptions, concerning in particular the descriptive concepts involved in association

with the objects they are used to describe. The basic claim is that the sheaf-theoretic descriptive framework makes good sense only on the basis of a *revised* realistic interpretation.

Our philosophical starting point is that a quantum event algebra reflects an *objective physical reality* at the quantum level having existence *independently* of some mind perceiving it. Evidently, such an interpretation viewpoint is compatible with the formalism of quantum theory if and only if a *non-Boolean* event structure is being manifested *globally*. Let us initially notice that we describe our *observations* using notions of validity adhering to *Boolean logic* as a consequence of the preparation of Boolean environments of measurement. This is due to the fact that only in such environments is it possible to *separate* elements and conceive of them as existing *in isolation* from the rest of the world. Indeed, the Boolean specification of environments of measurement incorporates the implicit assumption of a separative and compact underlying topology (Stone representation theorem).⁹ Of course, such an assumption, in general, can be used as a methodological tool of enquiry, *only locally or partially*, and can never be claimed to assume global validity in the name of empirical findings, being themselves amenable to the specification of such global environments. The appropriate border in the domain of validity of such a Boolean description is decided by the *relevant abstractions* of a measurement situation conceived precisely as a *Boolean localizing context*.

It is instructive at this point to remind ourselves of Bohr's definition of the word "*phenomenon*" to refer exclusively to observations obtained under specific circumstances, including the account of the whole experiment. Thus, for methodological purposes, we may adopt Bohr's concept of phenomenon as a referent of the assignment of an observable quantity to a system, legitimately thought of as approximately isolated, *strictly* in the context of a local Boolean environment of measurement. Let us notice that this consideration is naturally required because of the globally non-Boolean event structure, necessitating a *multitude* of locally considered, isolated yet correlated manifestations with respect to distinct Boolean reference frames. In this sense, every Boolean filtering context corresponds to a particular *type* of observable phenomenon and every proposition of the universe of discourse *belongs to* at least one Boolean frame. Thus every proposition is in principle descriptive of a classically conceived observable, but *only under* the choice of a local physical context. It is also important to notice that the physical claim of partial descriptions embodies in an essential manner the claim that the description of a phenomenon, as above, necessitates a theoretical conception of the phenomenon as the referent of an interactive process structured in *two levels*:

The first level of this process is constitutive of the *generation* of a *localizing environment*, as a reality probing filter, endowed with an intentionally prepared mechanism of abstraction determined explicitly by the qualitative observable nature of the specific Boolean environment (Boolean reference frame).

The second level of this process is constitutive of the *actualization* of some observed event only after *interaction* with the relevant measuring apparatus attached as a binary code to the corresponding Boolean frame.

The conception of phenomena in this approach enforces reference to quantum systems as actually existing objects behind the phenomena, the latter being only their *local or partial* manifestations with respect to Boolean environments designed for that purpose. Evidently, each separate phenomenon *cannot* be regarded as a full representation of a quantum system. Only at the level of sheaves of Boolean frames, that is *Boolean localization systems*, constituting consistently *interconnected multiplicities* of phenomena in terms of Boolean observable germs (equivalence classes), can an *isomorphic representation* of the structure of events of a quantum system be reproduced by means of the *inductive limit* construction.

In the light of the above, it is legitimate to say that the sheaf-theoretic localization scheme replaces the classical static monolithic realist view with a form of *relational realism*, admitting *multiplicities in logical space*, but not some “actualized” space of existing objects, as an expression of qualitative structured *potential continuous variation* in the observable universe of discourse, which are simultaneously strictly conditioned to obey *collectively* the *global constraint* of global uniqueness of a sheaf. Hence, this explicitly *rules out* any interpretation in the classical realist sense of a one-to-one correspondence between concepts used to describe a phenomenal object and the presumed properties of an independent reality. Still, a revised realistic understanding of the description of quantum systems is retained.

Consequently, by virtue of the sheaf-theoretic scheme of interpretation, the *classical realist assumption* that knowledge of an object is achieved by forming a representation of that object as a substance possessing properties is *rejected* and subsequently *replaced* by the possibility of formulating local or partially compatible contextual Boolean descriptions with respect to Boolean localization systems. The systems are then *grounded* on the same actually existing quantum object, where the sameness is precisely determined by the possibility of *uniquely gluing* these descriptions globally via the synthetic ontological process of sheaf-theoretic local-to-global formation. In this way, the revised realist position justifies the usage of the adjective *complex* when we refer to quantum systems. Moreover, they are distinguished from simple systems by applying precisely *the norm* of this revised realist position. So,

whereas classical systems are simple because they conform to the descriptive ideals of classical realism, quantum systems are *complex* because they conform to the more complex local-global requirements of revised relational realism.

In this way, we may argue that the quantum level of reality can be conceived only through a *relational sheaf-theoretic perspective*. It has now emerged that relations occupy substantial and remarkable territory in our description of nature, one which was once occupied exclusively by properties. An actually existing quantum system is *not* described through isolated properties, but only through its *relations* with localizing physical contexts interconnected together according to the *sheaf-theoretic* specifications of Boolean localization systems. More concretely, the topological covering process of a quantum event algebra associated with a Boolean localization system leads naturally to a contextual description of quantum events with respect to local Boolean reference frames of measurement. Stated more precisely, *quantum events are represented by evaluations of equivalence classes of Boolean coefficients (Boolean germs) with respect to local compatible Boolean contexts belonging to a Boolean localization system*. Thus, the global information content of a quantum event algebra remains *invariant* by the action of some topological covering system, if and only if that system forms a *Boolean localization system*. Hence, *the significance of a quantum event algebra is lifted from the orthoposet axiomatization at the level of events, to the sheaf-theoretic gluing conditions at the level of Boolean localization systems*. Eventually, the former axiomatization is fully and faithfully recaptured at the level of equivalence classes in these localization systems.

We argue that the relational sheaf-theoretic perspective on quantum reality provides a *viable mathematical model* to describe and understand Whitehead's conception of an *extensive event continuum*. Extension is fundamentally tied with the assumption of *potential divisibility* signifying a part-whole or local-global type of relation. In this way, the crucial role in the conceptualization of quantum event ontology is being played by the notion of a Boolean localization scheme, modeled in the *functorial /relational* terms of *sheaves*. The topological internal gluing relations characterizing sheaves force a consistency in the action of the inverse processes of extending from the local (Boolean) to the global (quantum) level and conversely reducing from the global to the local. Thus, *the real potentialities relative to all standpoints (local Boolean logical frames) are coordinated as diverse determinations of one extensive continuum, being a relational complex united by the sheaf-theoretic part-whole or local-global mereotopological relations*. Most importantly, the sheaf-theoretic extensive scheme provides the *generic morphology of the internal relations which bind the actual occasions into a nexus*. More concretely, the qualification of a uniform fibration expressing the potentialities relative to all Boolean frames, viz., of a presheaf of Boolean frames as a sheaf requires the explicit consideration of covering

sieves. The covering sieves are functorial objects acting as the semantic carriers of networks of internal relations among Boolean frames with respect to which the *process-theoretic scheme of extensive connection* in the continuum is consistently realized. Notice that the sheaf-theoretic scheme of extensive connection is based on the *ontological primacy of events* or facts or actual occasions. This is the case because each actual occasion *self-determines its own subjective standpoint* (Boolean logical frame), and simultaneously instantiates a *bundle of potential relations* referring to this standpoint. It is crucial that all these potential relations, viz., all the relations among local observables, are *induced* by the internal relations among their underlying standpoints. Now, the idea of an actual occasion conceived in the sense of an actualized *concrecence*, via the evaluation of some Boolean germ expressing its internal constitution in this process, serves as a *datum* for subsequent becoming actual occasions in the extensive connection of the continuum and so on. Consequently, a global quantum event continuum is *not* conceptualized as a completed a priori existing set-theoretic structure but it is constituted in a *continuous* topological process of extension from the local to the global by *actualization* of new potential facts with respect to local Boolean frames.

The framework of category theory in its function to express the process of extensive connection in the quantum event continuum via the notion of Boolean localization systems, that is in the topological terms of sheaf theory, is indispensable in order to understand the revised relational realist scheme in connection to Whitehead's process theory.

The most important aspect of its function in this context is the *conceptualization of the quantum event continuum via an inductive limit categorical construction in a sheaf of germs of local Boolean observable relations*. This is significant because the formulation of the mathematical procedure by which the *approximation to the global quantum level* is consistently achieved depends on the properties of the relation of *extension of Boolean germs* (equivalence classes), and most distinctively, this is implemented by *universal* means. This is the case because the inductive limit is described categorically via a *universal mapping property* and thus functions relationally as a *potentiality for actuality*.

Thus, from a generalized viewpoint, the congruence of the proposed sheaf-theoretic quantum relational realist position with Whitehead's process-theoretic scheme traces back to the semantic convergence between the category-theoretic notion of a *universal* (like the inductive limit) and Whitehead's notion of an *eternal object* or *pure potential for the specific determination of fact*. The theoretical basis of this semantic convergence stems from the fact that both of them act as *agents in objectification* under the operation of the *Category of Transmutation* which, in category-theoretic terms, is subsumed by the action of pairs of *adjoint functors*. In particular, keeping in focus quantum theory, the *mutually implicative dipolar functorial internal relation* between the multiplicity of local Boolean

frames and the global quantum is expressed in terms of a pair of adjoint functors between the category of sheaves of Boolean event algebras and the category of quantum event algebras. As we have argued, this categorical adjunction formalizes the process-theoretic operation of the category of transmutation by relating internally and bidirectionally the variable and local Boolean with the quantum global level of structure.

A remarkable application of this scheme is the theoretical explanation of *coarse-graining induced decoherence* and the *emergence* of classical behavior. In this way, coarse-graining induced decoherence is understood and modeled as an *intrinsic* reductive global-to-local process, in the sense that classical behavior emerges intrinsically from the global quantum level via coarsening the scale of resolution of quantum observable behavior by means of macroscopic Boolean observables locally according to the sheaf-theoretic topological compatibility relation.

Notes

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